

GUI Based ERROR CORRECTION IN DATA COMMUNICATION Using MATLAB

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Error correction and detection has great practical importance in maintaining data and ensuring that information is transferred intact from source to destination. Also, MATLAB based graphical user interface (GUI) for convolutional code (CC) provides basic concepts of CC, its encoder structure, and classification and decoding techniques.

CCs are widely used as channel codes in practical communication systems for error correction. Encoder design consists of current bits and previous bits that are stored in shift registers. Encoded output sequence is generated from present and previous message bits. Encoded output is passed to the

decoder and the decoding algorithm, which is used in Viterbi decoding algorithm.

Presented here is a project in which we design CCs for various code rates. The encoder of the designed CC is based on the code rate chosen by the user. Once the code is encoded, random noise can be added to the encoded sequence. The decoder is used to decode the message; at output, original message is obtained. Performance is evaluated with bit errors, which are the number of errors after binary sequence is decoded.

There are two major parameters on which CCs are based—code rate and constraint length. Code rate is defined as the number of current bits to the number of outputs, whereas constraint length is defined as the number of bits

stored in each shift register, including the current bit.

Note. In order to run this application, a 32- or 64-bit operating system with MATLAB software is needed. The designed GUI panel/program is user-friendly. Different buttons are used in this project, and the entire working of the project is performed through push buttons.

Block diagram of CC under different code rates, which are used in this application, is shown in Fig. 1. It illustrates the working of this project using MATLAB.

Software program

The application program is developed in MATLAB version R2013a. It has a GUI for CCs under different code rates. To open a GUI panel of the command guide, type on the command window.

When bit1.m program is run, a

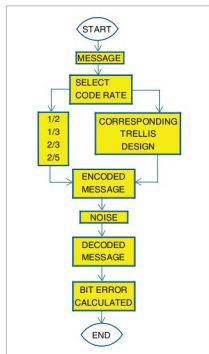


Fig. 1: Block diagram of convolutional code

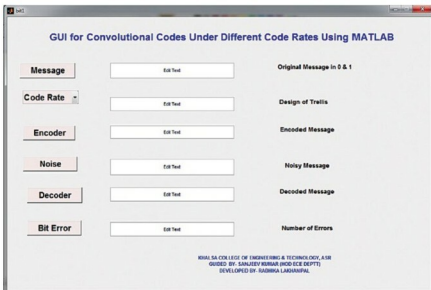


Fig. 2: Main GUI panel of the project